

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S. Adhi kesavan / 192521172(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

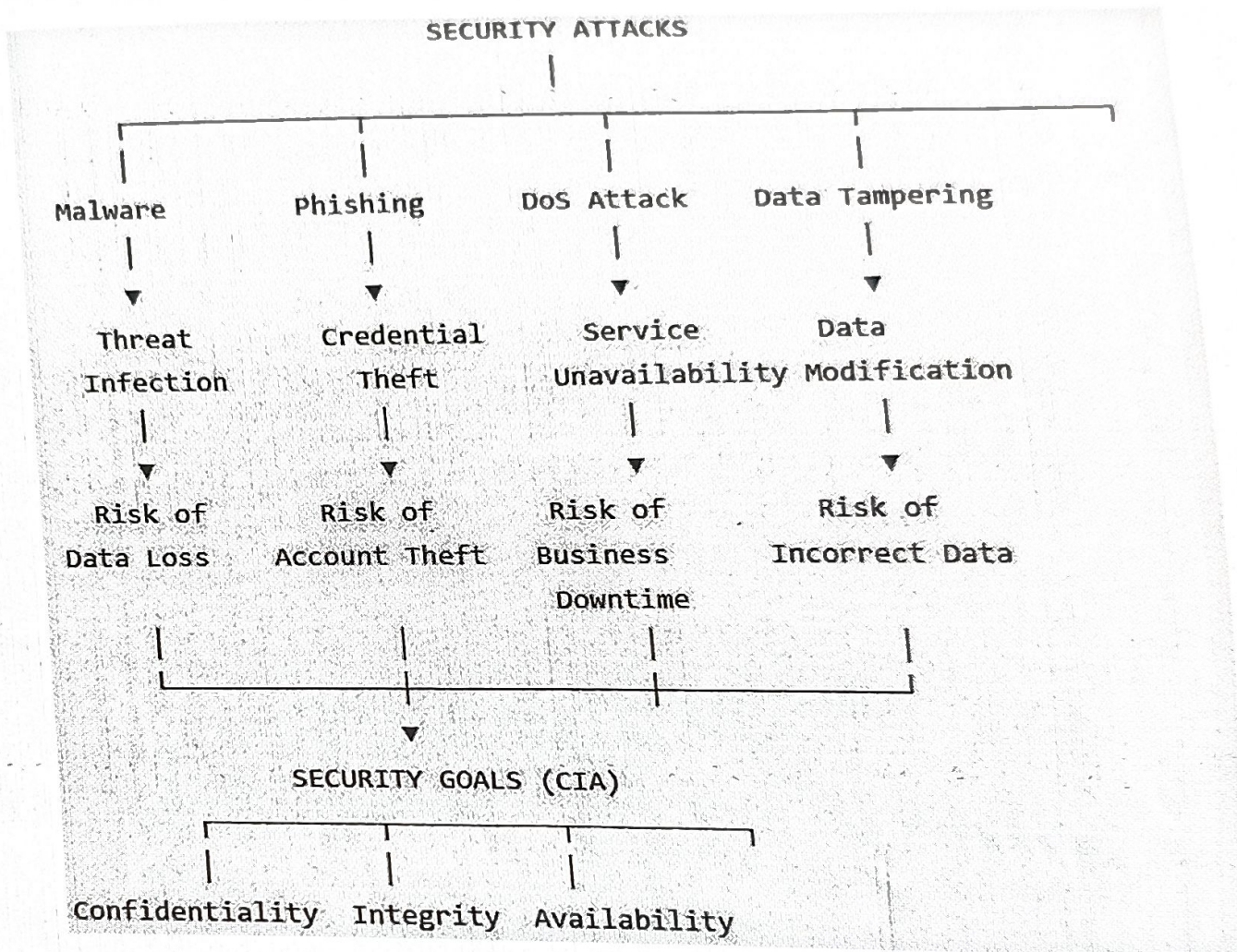
Signature of the student: Adhikesavan

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

Solution :

(1)



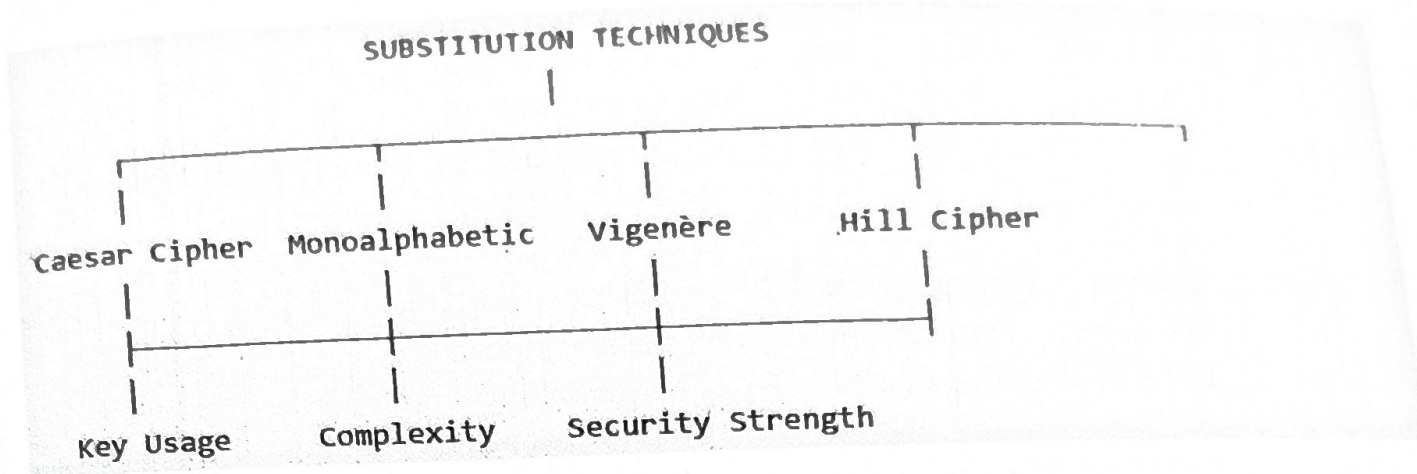
(a) Attack → Threat → Risk

Security Attack	Threat	Risk
Malware	System infection	Data loss, system damage
Phishing	Unauthorized access	Unauthorized account access
DoS (Denial of Service)	Website/server downtime	Website/server unavailable

(b) Security Goals and Mitigation

Security Goal	Mitigates	Example Protection
Confidentiality	Unauthorized access	Encryption, strong passwords
Integrity	Data modification	Hashing, Digital Signatures
Availability	Service interruption	Backup servers, Firewalls, Load Balancing

(2)



Comparison Based on Key Usage, Complexity and Security

Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Single shift key	Very Low	Very Weak
Monoalphabetic Cipher	One substitution alphabet	Low	Weak
Vigenère Cipher	Keyword (multiple shifts)	Medium	Moderate
Hill Cipher	Matrix key	High	Strong

Analysis: Most Secure Technique

Hill Cipher is the most secure among these four techniques.

Reasons

- Uses matrix multiplication instead of simple letter substitution.
- Encrypts multiple letters together (block cipher).
- Produces highly randomized ciphertext.
- Harder to break using frequency analysis.
- Requires knowledge of the matrix key for decryption.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation